

BahaAI — IoT Hardware Buying Guide

What to buy, how much it costs, and what it looks like

Flood Early-Warning System · Municipality of Consolacion, Cebu
Prepared 8 August 2026 · Budget build: **2 sets** — **approx. ₱3,900**

Read this first. Prices shown were checked on **Circuit Rocks (Philippines)** on 8 August 2026 and are the *ceiling* — the same modules on Shopee usually cost 30–50% less. Prices move, so treat every figure as a budgeting estimate, not a quotation.

Every item has a **yellow search box** with the exact words to type into Shopee or Lazada. That is the fastest way to find the part, and it keeps working after individual seller listings disappear.

About the pictures. These are hand-drawn illustrations, not photographs of specific products. They show each part's recognisable shape and colour so you can spot it in a listing. Real items vary in silkscreen colour and board layout — **match the part number in the search box, not the picture.**

1 What the software expects from your hardware

Your device only has to do one thing: send a small JSON message to the BahaAI server. Everything else — alerts, predictions, maps, notifications — already works.

Field	Rule	What it means for your hardware
<code>water_level</code>	Required · number ≥ 0	Water height in metres . This is the only field you must send. It comes from the ultrasonic sensor.
<code>rainfall</code>	Optional · number ≥ 0	Millimetres of rain. See the honest warning in Section 4.
<code>temperature</code>	Optional · number	Degrees Celsius — from the DHT22.
<code>humidity</code>	Optional · 0 to 100	Percent — from the same DHT22.
<code>recorded_at</code>	Optional · date	Leave blank and the server timestamps it on arrival.

Four rules that decide what you buy — do not skip these.

- 1. The sensor measures DOWN, the app wants UP.** An ultrasonic sensor on a post reports the *distance to the water*. The app wants the *water level*. Your code must do `level = mount_height - measured_distance`. Get this backwards and every alert fires inverted.
- 2. Minimum range is 25 cm.** The waterproof ultrasonic sensor is blind closer than 25 cm. You cannot mount it just above the water — leave at least 25 cm of air above the highest water you expect. Design your test tank around this.
- 3. Send a reading at least every 10 minutes.** The system marks a sensor "offline" after 30 minutes of silence and raises a fault alert. Use a **5-minute interval**.
- 4. The prediction needs at least 3 readings within 2 hours.** At 5-minute intervals you get 24 readings — comfortable. Any slower and the AI prediction silently returns nothing.

You only need ONE working sensor. The system handles 7 flood zones, but each is independent. One real device plus the built-in simulator (`php artisan sensors:simulate`) demonstrates the entire system. Do not buy seven of everything.

2 Set A — Bench & Testing Kit

For the desk, indoors, on a breadboard. This is where students write and debug the firmware before anything gets soldered or gets wet. **Subtotal** ≈ ₱1,444 (₱1,794 with the multimeter).



ESP32 Development Board (30-pin, WROOM-32) **MUST HAVE**

The brain. Reads the sensors, connects to Wi-Fi, and posts readings to BahaAI. Recognise it by the **silver metal shield can** and the **zigzag gold antenna** at one end. **Do not buy an Arduino Uno** — it has no Wi-Fi and cannot talk to the server.

₱178 – ₱355 Qty 1 per set

Search: "ESP32 development board 30 pin" or "ESP32 WROOM-32 devkit"



JSN-SR04T v2.0 Waterproof Ultrasonic Sensor **MUST HAVE**

Measures the water surface from above. **This is the single most important part — do not substitute a cheaper non-waterproof one for the real build.** Recognise it by the small blue board with a **separate silver cylinder on a cable**. Range 25 cm to 4.5 m.

₱250 – ₱557 Qty 1 per set

Search "JSN-SR04T" — also try "AJ-SR04M", the identical module under another name, often half the price.



HC-SR04 Ultrasonic (non-waterproof)

The cheap practice version — two silver "eyes" on a blue board. Use it to get the code working before you risk the expensive waterproof one. Buy two; they are almost free.

₱25 Qty 2

Search: "HC-SR04 ultrasonic sensor"

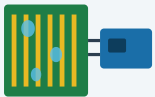


DHT22 Temperature & Humidity Sensor

Small **white plastic box with a grille** and 4 legs. Fills the **temperature** and **humidity** fields. The blue DHT11 (₱55) also works with slightly less accuracy.

₱109 Qty 1

Search: "DHT22 AM2302 sensor module"



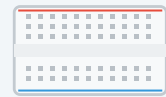
Rain / Steam Detection Module

READ SEC. 4

Green board with comb-shaped copper tracks plus a small blue driver board. **It only tells you wet or dry — it cannot measure millimetres.** See the warning in Section 4 before relying on it.

₱75 Qty 1

Search: "rain sensor module arduino"



Breadboard, 830 tie-point

White plastic board full of holes with red and blue rails. Lets students wire everything without soldering, then pull it apart and try again.

₱120 Qty 1

Search: "breadboard 830 point MB-102"



Jumper Wire Set (Male-Male, Male-Female, Female-Female)

Ribbons of coloured wires with plastic-shrouded pins. **Buy all three types** — running out of one kind mid-build is the most common way to lose an afternoon.

₱45 each Qty 3 sets

Search: "dupont jumper wires 40pcs male female"

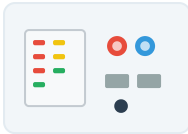


5V 2A USB Power Adapter + Micro-USB Cable

Buy a proper 2A adapter. An underpowered charger or a thin "charge-only" cable causes random ESP32 reboots that students will waste days blaming on their code.

₱180 Qty 1

Search: "5V 2A USB adapter" + "micro USB data cable"



Resistor / LED / Button Assortment Kit

Consumables for pull-up resistors, status LEDs and reset buttons. You will use these constantly and they are cheaper in a kit than singly.

₹125 Qty 1

Search: "resistor LED button breadboard kit"



Digital Multimeter OPTIONAL

Strongly recommended anyway. Without it, "is this wire actually connected?" and "is this pin really at 3.3 V?" become guesswork.

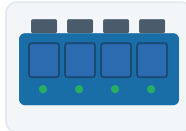
₹350 Qty 1

Search: "digital multimeter DT830B"

3 Set B — Prototype & Demo Kit

The presentable build: soldered, boxed, and able to physically pump water and sound an alarm in front of the panel. Includes a second ESP32 and ultrasonic so Set A stays intact for debugging. **Subtotal ≈ ₱2,435.**

Set B repeats four items from Set A — ESP32 (₱250), JSN-SR04T (₱400), DHT22 (₱109) and rain module (₱75). That is intentional: you do not want to dismantle a working demo every time someone needs to test a code change. Those four are listed in the cost table but not drawn again below.



4-Channel Relay Module, 5V **MUST HAVE**

Blue board with four blue cube-shaped relays. This is what lets the ESP32 switch the pump, siren and floodgate — the **Smart Actuation** screen in the app drives exactly these four channels.

₱150 – ₱316 Qty 1

Search: "4 channel relay module 5V optocoupler"



Mini Submersible Water Pump, 3-6V

Small plastic cylinder with a spout, sits in water. Represents the **drainage pump** actuator. At ₱41 it is the cheapest impressive thing in this entire list — it visibly moves water when an alert fires.

₱41 Qty 1

Search: "mini submersible water pump 3-6V"

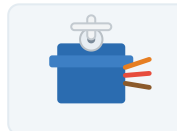


Active Buzzer / Piezo Siren, 5V

Small black cylinder. Represents the **warning siren** actuator. Get the "active" type — it beeps from plain DC power with no extra code.

₱30 Qty 1

Search: "active buzzer 5V arduino"

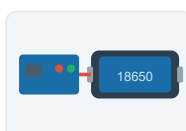


SG90 Micro Servo Motor

Small blue box with a white plastic arm. Represents the **floodgate / control valve** actuator — it physically swings a miniature gate open and shut on command.

₱120 Qty 1

Search: "SG90 micro servo motor 9g"



TP4056 Charger + 18650 Cell + Holder

Battery backup so the node keeps reporting during a brownout — a genuinely defensible design point for a flood system in the Philippines. Small blue charging board plus a AA-looking-but-larger cell.

₱280 Qty 1 set

Search: "TP4056 charging module" + "18650 battery holder"

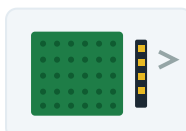


IP65 Weatherproof Junction Box

Grey sealed box with corner screws. **Buy this at a hardware store, not an electronics shop** — Ace, Wilcon or your local hardware will be roughly half the price for the identical item.

₱250 Qty 1

Search: "IP65 junction box outdoor electrical"



Perfboard + Pin Headers + Solder

Once the breadboard version works, transfer it to perfboard so the demo does not fall apart when someone bumps the table. Green board of holes plus black header strips.

₱200 Qty 1 set

Search: "perfboard prototype PCB" + "male pin header 40 pin"



Test Tank — Clear Container / PVC Pipe + Tubing

Your artificial river. A tall transparent container lets the panel **see** the water rise past the warning and critical lines while the dashboard reacts. Remember the 25 cm minimum sensor gap.

₱350 Qty 1

Hardware store: tall clear acrylic/plastic container, 4" PVC pipe, 6 mm silicone tubing

4 The one honest compromise — rainfall

Rainfall is the only field where the cheap part cannot do the real job. Decide this deliberately and write your decision into the paper.

The ₱75 rain module cannot measure millimetres. It detects whether its surface is *wet or dry*. If you feed its output into the `rainfall` field, you are publishing an invented number — and a panel member who knows sensors will ask. Use it only as a "raining right now" indicator, and say so explicitly.

Option	Cost	Trade-off
A. Use the weather API Recommended	₱0	BahaAI already pulls rainfall for Consolacion from OpenWeatherMap. Leave <code>rainfall</code> empty on the device and let the weather service supply it. Defend it as deliberate separation of data sources — water level from your sensor, meteorology from a meteorological service. Costs nothing and is fully honest.
B. Build a tipping bucket Best for marks	≈ ₱150	A reed switch (₱30) plus a magnet and a small see-saw bucket you 3D-print or cut from a bottle. Calibrate by pouring a measured volume and counting tips. Real millimetres, and it is an original contribution you can write a whole chapter about.
C. Buy a tipping bucket	₱1,200 – 2,500	Accurate and finished, but it is the most expensive single item on this list — for a field you marked optional. Only justified if the study is specifically <i>about</i> rainfall measurement.
D. ₱75 wet/dry module	₱75	Indicator only. Fine as a supplementary "it is raining" flag. Never send it as a millimetre value.

5 Complete cost summary

Item	Set A	Set B	Unit ₱	Total ₱
ESP32 Development Board	1	1	250	500
JSN-SR04T Waterproof Ultrasonic	1	1	400	800
HC-SR04 Ultrasonic (practice)	2	—	25	50
DHT22 Temp/Humidity Sensor	1	1	109	218
Rain / Steam Module	1	1	75	150
Breadboard 830-point	1	—	120	120
Jumper wire sets (3 types)	3	—	45	135
5V 2A adapter + micro-USB cable	1	1	180	360
Resistor / LED / button kit	1	—	125	125
4-Channel Relay Module 5V	—	1	150	150
Mini submersible pump 3–6V	—	1	41	41
Active buzzer / siren 5V	—	1	30	30
SG90 micro servo	—	1	120	120
TP4056 + 18650 cell + holder	—	1	280	280
IP65 junction box	—	1	250	250
Perfboard + headers + solder	—	1	200	200

Test tank + tubing	—	1	350	350
Core subtotal — both sets				3,879
Digital multimeter <i>(optional but advised)</i>	1	—	350	350
Spares — 1 extra ESP32 + jumpers <i>(advised)</i>	—	—	300	300
DIY tipping bucket rain gauge <i>(Option B)</i>	—	1	150	150
RECOMMENDED TOTAL				₱4,679

Absolute minimum viable build: ₱1,444 — Set A alone with one ESP32 and one waterproof ultrasonic proves the entire data path end to end. Everything in Set B makes the defense more convincing; none of it is required for the system to work.

6 What you do NOT need to buy

Item	Money saved	Why you can skip it
GSM / SIM800L module	₱500+	The demo runs on a phone hotspot. Mention cellular as future work for genuine field deployment.
Paid SMS gateway credits	₱500+	The system already logs SMS as simulated while sending real email. If your panel insists, Semaphore PH is about ₱0.50 per message.
Raspberry Pi	₱2,500+	Five times the price of an ESP32 and heavy overkill for reading three sensors.
A second complete sensor node	₱1,700	Two nodes demonstrate nothing that one node plus the built-in simulator does not already show.
Solar panel	₱785+	Nice story, poor value for a demo. The 18650 battery already covers the "keeps running in a blackout" narrative.

7 Where to buy

Circuit Rocks **VERIFIED**

circuit.rocks

Manila-based, same-day delivery in Metro Manila. Every peso price in this guide was checked here on 8 Aug 2026. Higher prices than Shopee but you get the right part, quickly.

Best for: the JSN-SR04T when you cannot afford a wrong order.

Shopee PH **CHEAPEST**

shopee.ph

Typically 30–50% cheaper for identical clone modules. Delivery 3–7 days. Check seller ratings and that the shop is **Philippines-based**, not an overseas listing.

Best for: everything except parts you need this week.

Lazada PH **BACKUP**

lazada.com.ph

Similar pricing to Shopee, sometimes better stock on relays and power supplies. Use LazMall sellers for faster, more reliable shipping.

Best for: when Shopee is out of stock.

Makerlab Electronics

makerlab.ph — one-year local warranty and real technical support. Higher prices, but the place to go for the tipping-bucket rain gauge if you choose Option C.

e-Gizmo Mechatronics

Sta. Mesa, Manila — walk-in. If students are in Metro Manila, one afternoon here beats a week of waiting for deliveries. Bring this printout.

Local hardware store

For the **junction box, container, PVC pipe and tubing**. Roughly half the price of the same items from an electronics shop.

Avoid Temu for this project. Lead times run two to four weeks and you cannot return a dead sensor in time for a defense deadline. The saving is not worth the schedule risk.

Two buying tips. Always search "**AJ-SR04M**" as well as "**JSN-SR04T**" — same module, different name, often half the price. And check the seller is shipping from within the Philippines; overseas listings on Shopee quietly add two weeks.

8 Printable shopping checklist

Tick items as they arrive. Test each part the day it arrives — a dead module discovered three weeks later cannot be returned.

Item	Qty	Budget ₱	Received / tested
☐ ESP32 development board	2	500	
☐ JSN-SR04T / AJ-SR04M waterproof ultrasonic	2	800	
☐ HC-SR04 ultrasonic (practice)	2	50	
☐ DHT22 temperature / humidity sensor	2	218	
☐ Rain / steam module	2	150	
☐ Breadboard 830-point	1	120	
☐ Jumper wires — M-M, M-F, F-F	3	135	
☐ 5V 2A adapter + micro-USB data cable	2	360	
☐ Resistor / LED / button kit	1	125	
☐ 4-channel relay module 5V	1	150	
☐ Mini submersible pump 3–6V	1	41	
☐ Active buzzer 5V	1	30	
☐ SG90 micro servo	1	120	
☐ TP4056 + 18650 cell + holder	1	280	
☐ IP65 junction box	1	250	

☐	Perfboard + headers + solder	1	200
☐	Test tank + tubing	1	350
☐	Digital multimeter <i>(optional)</i>	1	350
☐	Spare ESP32 + jumpers <i>(advised)</i>	1	300
☐	DIY tipping bucket parts <i>(optional)</i>	1	150
TOTAL			₱4,679

9 Two software changes needed before hardware works

1. The reading endpoint has no authentication. `POST /api/sensors/{id}/readings` is currently open — anyone who knows the address can inject fake readings and trigger real alerts to real residents. Acceptable on a closed bench network; **add a Sanctum API token before this touches a public network**, and document the mitigation in the paper.

2. Devices cannot report their battery level. The `sensors` table has a `battery_level` column and the dashboard displays it, but it is not accepted by the reading endpoint — so a real node has no way to send it. If you want that panel to show live data, add `battery_level` to the request validation rules and to the API controller.